DATATHON PROTOTYPE · DEMONSTRATED ONLY · NOT YET VALIDATED

2026 NC GLOBAL HEALTH CONFERENCE

From Surveillance to Personal Health Insight

Using Generative AI to Rapidly Prototype People-Centered Public Health Tools

William Powell, Yomna Abdelazim, John Hanna, Tyler Humpherys, Cristina Lee, Vivian Muzyk, Jeslyn Pratiknjo, Aitong Ruan · UNC Chapel Hill

This research originated at the 2026 “UNC Gillings AI and Public Health Showcase and Datathon,” an event hosted by the UNC Gillings Center for Artificial Intelligence and Public Health (CAIPH).

Gillings School of Global Public Health

App code and coefficients
github.com/liampwl/gillings-datathon

356,158
adults answered BRFSS in 2025; none got anything back. Our model used 2015 data.

2 days
from a public dataset to a working prototype, built with AI coding agents.

5 changes
to move from a two-day demo to a pilot, listed in a review of the code.

01 Background and methods

The Behavioral Risk Factor Surveillance System (BRFSS) is the largest continuously conducted health survey in the world, covering all 50 states. Respondents learn nothing about their own health from taking it.

Could a surveillance survey return something useful to the people who answer it, without changing the data it collects?

Setting. An eight-person team at the 2026 UNC Gillings AI and Public Health Showcase and Datathon, March 6–7, 2026.

Outcome. Self-reported diabetes or prediabetes in the BRFSS 2015 Diabetes Health Indicators extract, with 21 predictors. This is reported diagnosis at one point in time, not future risk, and undiagnosed cases count as negative.

Models. Logistic regression, random forest and XGBoost, each trained on a 50/50 balanced sample and on the original distribution, evaluated on one held-out split (sizes not recorded). ROC AUC was 0.82–0.83 for all six; logistic regression was kept for its readable coefficients. The extract has no survey weights, so none were used.

Development. AI coding agents (GitHub Copilot; Claude and Gemini in Google Antigravity) wrote the survey, scoring service and results pages. Team members wrote the prompts, tested on their phones and chose what to keep.

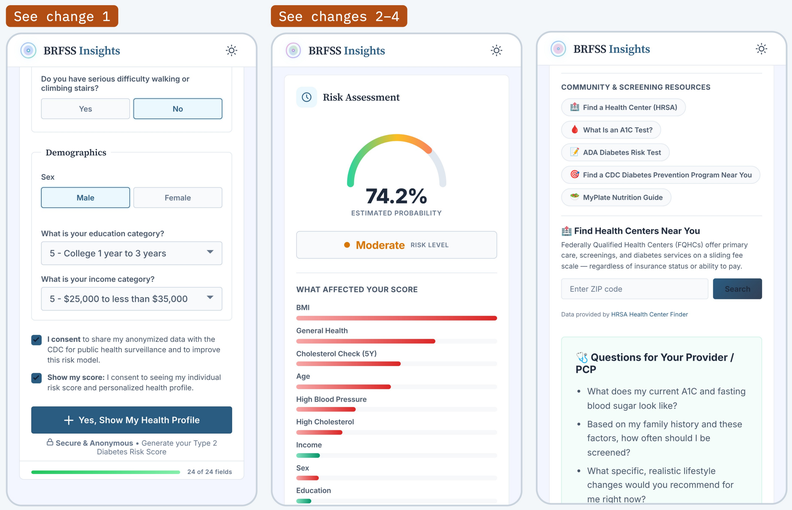
Review. In October 2026 the first author reviewed the final code, interface text and model outputs, with AI assistance. This was one reviewer's check, not a formal audit.

What a respondent would see

Answer 21 BRFSS questions → **See an estimate and what drove it** → **Get screening links and nearby clinics**

02 Prototype

See change 1 · See changes 2–4



Consent and opt-in · **Score and what affected it** · **Next steps and local resources**

Screens as built at the datathon, before the changes in section 03. Responses are hypothetical.

Design features

Separate opt-in. Seeing a score is a choice made apart from agreeing to data use.

Plain-language results. Participants enter age, height and weight instead of BRFSS codes, and each factor that moved the estimate is explained.

Next steps. CDC, ADA and NIH screening resources, a ZIP code search for federally qualified health centers (HRSA), and questions to bring to a provider.

Built to demo. Hosted on Vercel, so anyone at our presentation could open it from a link.

Data handling

Audience members at our datathon presentation tried it with whatever answers they liked. The scoring service has no database and does not store or forward responses. The only outside request is the optional ZIP code, sent to the HRSA health-center locator.

03 From demo to pilot

A two-day demo takes shortcuts. Reviewing the code afterward, we listed what would change before it is used with real respondents.

1 Write consent with survey staff
The demo’s consent box used placeholder wording saying answers would go to the CDC. Nothing was sent anywhere, and in voice mode the box checked itself.

2 Say what the number means
The model estimates the odds of a reported diabetes or prediabetes diagnosis, of either type. The demo labeled it a future type 2 risk score.

3 Ship the model the team reported
The app picked up the 50/50 training version instead of the full-extract model the team reported, so its percentages run high: 49% versus about 13% for the same average answers.

4 Drop the what-if simulator
It let people change answers and watch the score move, but these coefficients are associations, not effects, and two run against clinical expectation.
`CholCheck β = +1.31 · HvyAlcoholConsump β = −0.74`

5 Finish the demo stand-ins
Use one set of risk cutoffs across screens, and replace the placeholder neighborhood score with Social Vulnerability Index data.

Conclusion

In two days, AI coding agents let a team without dedicated developers build a working demo that gives BRFSS respondents something back. The changes above are the kind a fast build skips and an expert review catches, so that review must be a planned step before tools like this reach the public.

04 Next steps

We shared the prototype with NC DHHS in May 2026 and are exploring a possible pilot collaboration.

Keep feedback outside the survey

Offer it as an optional module after the interview is complete. The survey questions would stay the same, but advertising feedback in advance could change who responds and how, so measurement would need to be checked, not assumed.

Use a validated instrument

Replace the datathon model with the CDC/ADA prediabetes risk test. It needs items the datathon model did not use, including family history and gestational diabetes, and gives a points score, with 5 or more meaning increased risk. It does not give a probability.

Evaluate with respondents and survey staff

Measure comprehension, whether participation feels voluntary, accessibility, usefulness, anxiety or false reassurance, accuracy of resource links, and effects on survey completion and answers.